

FOCUS - Facilitating low-resolution OCR Confirmation with Unsupervised Systems

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<https://github.com/Shikhar2004-prog/MBCC>

Abstract. Optical Character Recognition (OCR) is the process of converting document images into text. Verification of OCR text (or confirmation) for low-resolution inputs remains challenging due to semantic and structural confusion between the input image and text predictions. The absence of labels in unsupervised settings further increases the complexity. We propose an unsupervised technique as a baseline for the verification of a hundred low-resolution images and their predictions from Tesseract OCR. To be precise, we first synthesize the images of OCR predictions using a fixed font and style. Next, we feed the low-resolution (real) input images and the corresponding (predictions') synthetic images to ResNet50 to extract word-level features. We finally compare the obtained features for verification using cosine similarity. To benchmark, we also present the results of Vision Language Models (VLMs) like ChatGPT, Claude, and Gemini by prompting them to compare low-resolution images to the predictions' synthetic images. We also present an ensembling approach and compare it with human evaluation. The work contributes to a broader understanding of low-resolution OCR in scenarios with a scarcity of human annotators.

Keywords: Unsupervised OCR, OCR text verification · low-resolution OCR · ResNet50 · Large Language Models · Text Image Verification

1 Introduction

The process of transforming the document images into text is called Optical Character Recognition (OCR). OCR text verification at low resolution is a challenging research problem due to the difficulty of capturing semantic and structural comparisons between OCR inputs and predictions. Low-resolution image verification finds use in document authenticity, archive digitization, and information retrieval from scanned documents among other things. To further underline the pragmatic importance of this study, we have included in the Introduction section actual application cases including historical document digitalization, forensic document verification, and low-resource archive indexing in government and educational settings. [13]. The unsupervised version of the verification task for low-resolution OCR text is further challenging due to the absence of labels.

Traditional OCR methods work well with images with normal or high resolution (≥ 150 dots per inch or dpi), but they have trouble with low-resolution images (especially the ones with ≤ 100 dpi) [10,3]. The limitation becomes even more problematic when resources are limited and high-resolution scanning capabilities are unavailable, or when working with historical documents or images captured under critical conditions, as noted by Springmann et al. [13]. Furthermore, different font styles, text layouts, and degradation patterns create extra problems that make verification even less accurate.

Pre-trained Convolutional Neural Networks (CNNs) like Resnet50 are efficient at extracting features from low-resolution images [4]. Moreover, the large language models with vision capabilities

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has shown promise in understanding text from image data [9]. Verification of OCR text using unsupervised approaches has garnered a lot of attention recently. Both Peng et al. [8] and Li et al. [7] have developed a transformer-based framework for OCR. Peng et al. [8] propose a self-supervised technique for document image representation learning. When it comes to capturing the structural relationships that exist between text parts, end-to-end techniques have demonstrated remarkable results in the normal settings [14]. OCR for low-resolution images offers issues that have been solved using a variety of approaches. A reading text in the wild approach was presented by Jaderberg et al. [6] with the purpose of enhancing the capabilities of text recognition. The performance of lower resolution machine recognition capabilities has been investigated by Gibley et al. [3], thus stressing the need for enhanced computational approaches through their findings.

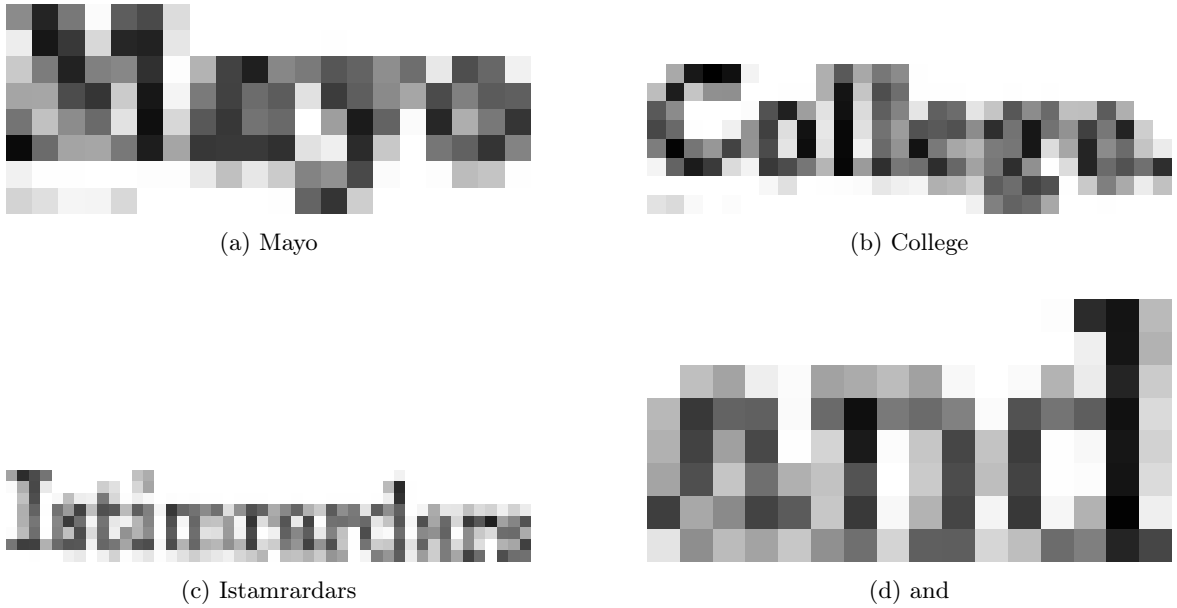


Fig. 1: Examples of low-resolution images used in our work. Below each image we present its label.

We propose a strategy that leverages the Tesseract for initial OCR text predictions, followed by word synthesis of predicted text with a fixed font style. We then perform feature extraction for the real input image (see samples in Fig. 1) and the corresponding synthetic image (of prediction) using the ResNet50 model. This approach helps to narrow the gap in unsupervised text image verification at low-resolution.

The key contributions of our work are:

1. We present our work for the OCR text verification for low-resolution images and a simple unsupervised baseline for benchmarking.
2. We develop a comprehensive benchmark for comparing the performance of different Vision Language Models (VLMs) and humans in low-resolution OCR text verification tasks.

2 Related Work

Optical character recognition (OCR) technologies, deep learning models, and human-computer interaction approaches have been employed to conduct extensive research on text recognition in printed document images. This section summarizes key contributions that have inspired our methodology.

2.1 Low-Resolution OCR Challenges

High-resolution images have traditionally been the primary focus of optical character recognition systems [12]. An initial method for executing OCR on low-resolution text images is introduced by Jacobs et al. [5], specifically related to images obtained from low-resolution cameras. Gilbey et al. [3] discuss significant degradations in OCR accuracy when processing documents below 100 dots per inch (dpi), with performance declining proportionally with resolution. While these works highlight challenges in low-resolution OCR, none of them work on the unsupervised method for verification of low-resolution OCR predictions.

In the domain of human-machine performance comparison, Schenkel et al. [11] conduct comprehensive studies comparing human and machine text recognition capabilities under varying conditions, concluding that humans demonstrate superior recognition abilities at lower resolutions. Similar to Schenkel et al. [11], we also compare machine and human performance, but for the low-resolution OCR text verification task.

2.2 Unsupervised OCR Methods

When labeled data is scarce, unsupervised methods for OCR prove particularly valuable. Aberdam et al. [1] develop a self-supervised framework for text recognition that operates independently of labeled data. Peng et al. [8] improve the area of unsupervised OCR by creating a contrastive learning framework specifically designed for document image analysis. Although there are several works on self-supervised and unsupervised OCR [8,15,1], none of them works in an unsupervised OCR environment at lower resolution images.

3 Methodology

This section outlines our methodology for unsupervised text verification with a primary focus on text images at the low-resolution of 75 dots per inch (dpi). The methodology has several important parts, such as preparing the dataset, extracting text using Tesseract, creating synthetic words, extracting features using the ResNet50 CNN model, and setting up a framework for evaluation by a large language model and a human.

3.1 Dataset Preparation

The data set consists of a hundred word images extracted from five publicly available documents, two scientific papers, and six books. Fig. 1 and Fig. illustrate examples of the some word at low-resolution levels (75 dpi).

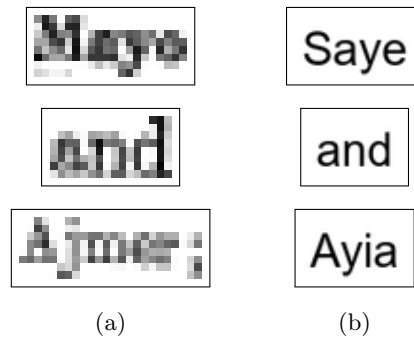


Fig. 2: (a) Shows the real text images at a lower resolution, and (b) displays the corresponding synthetic text renderings generated from the Tesseract OCR prediction with a fixed font and style.

3.2 Text Extraction with Tesseract OCR

To extract text from word images with low resolution, we made use of Tesseract, a Google optical character recognition (OCR) engine. We are able to adjust the engine parameters for our particular application since the text extraction method was built particularly for word-level recognition rather than document-level processing.

3.3 Synthetic Word Generation

Inspired by the approach of Etter et al. [2], we developed a synthetic word generation pipeline to produce standardised word images of the extracted text. This strategy addresses challenges related to font diversity, text styles, and multilingual complexity by enabling consistent feature comparison across varied real-world image texts.

As illustrated in Figure 2, the original text images (a) suffers from low resolution, noise, and inconsistent styling, making direct feature comparison challenging. By using Tesseract OCR to generate text predictions and rendering them with consistent parameters (b), we create normalised representations that preserve the semantic content while eliminating style variations, resulting in more effective feature matching during retrieval.

The synthetic image generation pipeline comprises the following steps:

1. **Text normalization** by removing unwanted characters such as double quotes from the input labels.
2. **Rendering using a consistent font**, specifically Arial, at a fixed font size.
3. **Precise layout control**, where each word is rendered onto a white background with uniform padding.
4. **Resolution standardization**: Each rendered image is resized to a height of 32 pixels while preserving its aspect ratio, enabling uniformity for downstream processing.

3.4 Feature Extraction with ResNet50

Feature Extraction with ResNet50

To extract discriminative visual features from both synthetic and real word images, we utilize the ResNet50 architecture [4], pre-trained on ImageNet. Rather than using the final classification layer, we extract feature embeddings from an intermediate convolutional layer, specifically `conv4_block6_out`. This layer offers a trade-off between low-level texture features and high-level semantic abstraction, making it especially effective for fine-grained word-level image matching tasks. Prior studies have shown that mid-level convolutional layers in deep CNNs retain spatial and structural information crucial for comparing local patterns [16], which is particularly important in distinguishing subtle variations in character shapes and word structures.

Images are first resized to a standard input resolution of 224×224 and normalized to align with the model’s expected input distribution. The resulting feature maps are globally averaged to produce fixed-size vector embeddings. A pair of images is classified as *Matched* if the similarity exceeds a chosen threshold; otherwise, it is labeled as *Mismatched*. This mid-layer-based extraction strategy enhances generalization and robustness in visually weak or low-resolution word images.

3.5 Similarity-Based Classification Using Mid-Level CNN Features

These embeddings are then compared using cosine similarity, a widely used metric for high-dimensional feature comparison due to its scale-invariant properties. The verification process determines whether a given text image corresponds to its purported textual content. Our approach computes the cosine similarity between the feature vectors of the original text image and its synthetically rendered images:

$$\text{similarity}(I_{orig}, I_{synth}) = \frac{\mathbf{f}_{orig} \cdot \mathbf{f}_{synth}}{\|\mathbf{f}_{orig}\| \times \|\mathbf{f}_{synth}\|} \quad (1)$$

where $\mathbf{f}_{orig}$ and $\mathbf{f}_{synth}$ are the feature vectors extracted from the original and synthetic images, respectively.

The threshold $\tau=0.48$ was determined empirically through validation experiments to optimize the trade-off between false accepts and false rejects.

3.6 Evaluation Framework

To evaluate our approach, we developed a multi-faceted assessment framework that includes:

1. **Human vs. machine performance:** Following Schenkel et al. [11], we conducted experiments comparing human recognition capabilities with our automated approach at the challenging 75 dpi resolution.
2. **LLM and Vision-Language Model evaluation:** We assessed the performance of state-of-the-art models with visual capabilities on our text verification tasks, implementing a standardized evaluation protocol.
3. **Error analysis:** We categorized verification errors into specific types (e.g., character substitutions, insertions, deletions) to identify systematic patterns and improvement opportunities.

Predictions are evaluated against ground-truth annotations using standard classification metrics—accuracy, precision, recall, and F1-score—to assess the effectiveness of the extracted features in modeling visual similarity.

4 Experimental Results

4.1 Performance Comparison

We conducted a comprehensive evaluation of text image verification by comparing human performance against state-of-the-art large language models (LLM) and vision language models (VLMs). The primary focus was on the critical threshold of 75 dpi, where human verification remains relatively reliable while automated systems begin to struggle significantly. For each system, we measured standard metrics, including accuracy, precision, recall, and F1 scores.

Table 1: Performance Comparison at low-resolution

System	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)
Our Baseline	49.00	30.77	76.92	43.96
Gemini	90.00	83.33	76.92	80.00
Claude	91.00	83.33	80.00	81.63
ChatGPT-4	92.00	87.50	80.77	84.00
Human Evaluators	95.00	95.83	85.19	90.20
Majority Voting	97.00	95.83	92.00	93.88

Performance Analysis As expected, the performance gap narrows considerably at the resolution of 75 dpi, as shown in Table 1. The data indicates that ChatGPT-4 achieves the accuracy and F1 score of 92.00% and 84%, significantly surpassing the baseline accuracy and F1 score of 49% and 43.96%. Yet ChatGPT-4 trails human performance with an average F1 score of 90.20 (averaged over 3 humans). The result suggests that at 75 dpi characters are distinguishable by humans more than Vision Language Models (VLMs). At this resolution, VLMs, including ChatGPT-4, achieve comparable accuracy to humans, though there is a notable 6.20% difference in F1 scores compared to human performance. An ensembling approach including majority voting from different VLMS and humans further improves the F1 score by 3.68% as shown in the last two rows of Table 1. This shows the effectiveness of human-in-the-loop in addition to VLMs.

4.2 Analysis of Results

The experimental results demonstrate several key findings:

1. **Critical dpi Threshold:** The 75 dpi level represents a significant threshold where human perception maintains high reliability while automated systems experience a reduced in performance.
2. **Model Ensembling:** Among the LLMs and VLMs tested, ensembled mode consistently outperformed other methods, probably because of its using all capabilities of other methods .

4.3 Comparative Analysis of Human and LLM Performance

To evaluate the effectiveness of both human evaluators and LLMs in verifying low-resolution OCR outputs, we analysed their performance across four scenarios. Figures 3, 4, 5, and 6 illustrate qualitative examples across different agreement categories between humans and LLMs.

Success by Both Humans and LLMs In several cases, both humans and LLMs correctly verified matches or mismatches. These examples typically retained sufficient visual clarity despite low resolution.

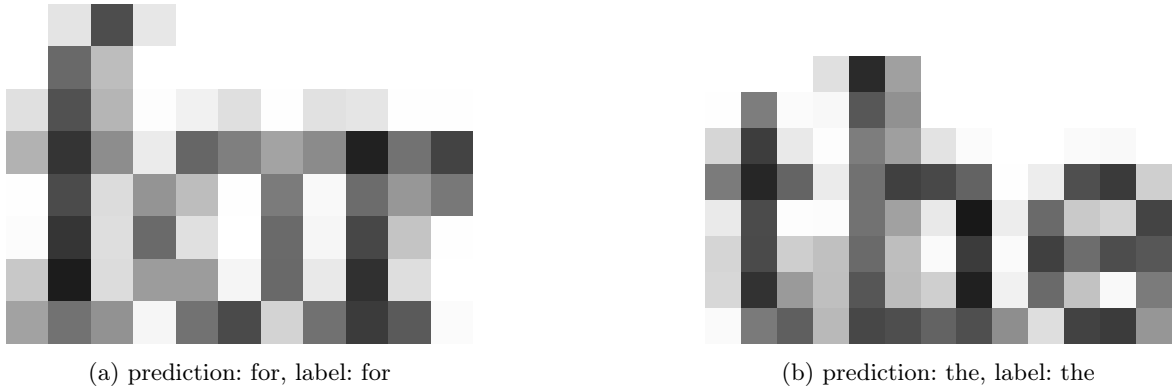


Fig. 3: Cases where both LLMs and human evaluators succeeded.

Human Success, LLM Failure Some instances were easily recognised by humans but misclassified by LLMs, indicating the limitation of current VLMs in subtle or degraded conditions.



Fig. 4: Cases where humans succeeded but LLMs failed.

LLM Success, Human Failure Conversely, LLMs occasionally succeeded where human evaluators erred, showcasing the robustness of VLMs under some noisy conditions.

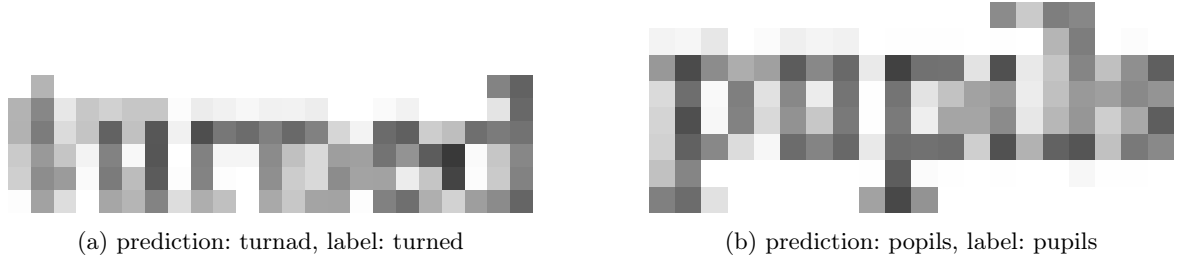


Fig. 5: Cases where LLMs succeeded but humans failed.

Failure by Both Humans and LLMs Finally, there are challenging cases where both approaches failed, typically due to extreme degradation or ambiguous character structures.

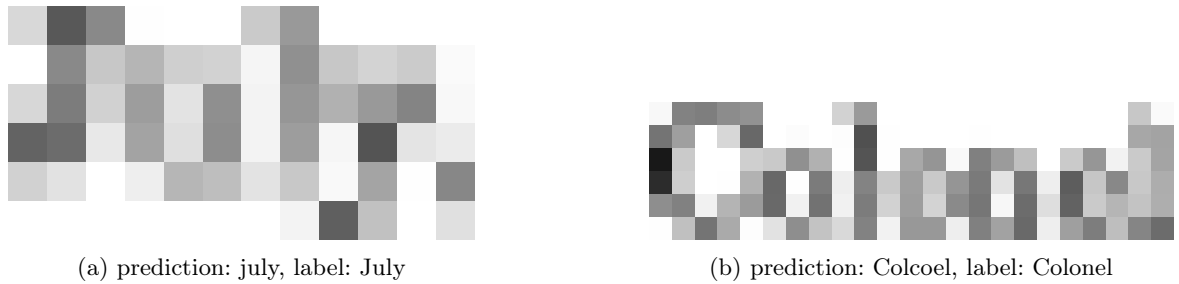


Fig. 6: Cases where both LLMs and humans failed.

5 Conclusion and Future Work

Our findings indicate that human evaluators consistently surpass AI models at lower resolutions, highlighting the substantial limitations of current AI and large language models in challenging visual contexts. At 75 dpi, ChatGPT-4, Claude, and Gemini attained an accuracy exceeding 90% and an F1 score exceeding 80, while our baseline model achieved only an accuracy exceeding 49% and a 43.96% F1 score. The gap highlights the persistent challenge of creating automated systems that can mimic human perceptual abilities in low-resolution OCR contexts. Future research aims to improve baseline performance by incorporating semantic and structural relationships among text characters, with a specific focus on enhancing resolutions below 75 DPI. This approach seeks to close the performance gap between humans and AI, enhancing automated visual text verification systems, besides that we plan to explore the transformer-based architectures, such as Vision transformers(Vits), which have shown promise in modeling long range dependencies and these models enable more features particularly for visually degraded diverse text images.

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